



Find Duplicate Characters Using Bit Manipulation

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Given a string containing lowercase English letters, find and print all the characters that occur more than once.

You must use bit manipulation to track the characters that have already appeared.

Input Format

The first line contains a string S

Constraints

$1 \leq |S| \leq 10^5$ S contains only lowercase English letters (a-z)

Output Format

Print all duplicate characters in the order of their first repeated occurrence. If there are no duplicates, print No duplicates.

Sample Input 0

```
programming
```

Sample Output 0

```
r g m
```

Sample Input 1

```
greater
```

Sample Output 1

```
r e
```

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Language

Python 3



```
1 import sys
2
3 def find_duplicates():
4     input_data = sys.stdin.read().split()
5     if not input_data:
6         return
7
8     s = input_data[0]
9
10    seen_mask = 0
11    duplicate_mask = 0
```

```
12
13     for char in s:
14         bit_index = ord(char) - ord('a')
15         if (seen_mask & (1 << bit_index)) != 0:
16             duplicate_mask |= (1 << bit_index)
17         else:
18             seen_mask |= (1 << bit_index)
19
20     if duplicate_mask == 0:
21         print("No duplicates")
22         return
23
24     printed_mask = 0
25     duplicates = []
26
27
28     for char in s:
29         bit_index = ord(char) - ord('a')
30
```

Line: 40 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

```
1 | programming
```

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✓ Sample Test case 1

Your Output (stdout)

```
1 | r g m
```

Expected Output

```
1 | r g m
```

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